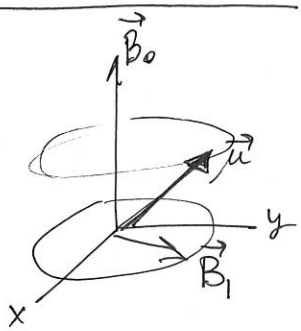
Magnetic Resonance



$\vec{B}_0$ is static.

$\vec{B}_1 = B_1(\cos\omega t, \sin\omega t, 0)$

rotates in xy-plane.

$B_1 \ll B_0$.

Expect <u>resonance</u> when $\omega \approx \omega_0$.

Assume $\mu_\perp \ll \mu_z \approx \mu$.

Then $\vec{\mu} \times \vec{B}_1 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \mu_x & \mu_y & \mu_z \\ \cos\omega t & \sin\omega t & 0 \end{vmatrix} B_1 = B_1\Big(-\hat{x}\,\mu_z \sin\omega t + \hat{y}\,\mu_z \cos\omega t$
$+ \hat{z}\,[\mu_x \sin\omega t - \mu_y \cos\omega t]\Big)$

But $\mu_{x,y} \ll \mu_z \Rightarrow \vec{\mu} \times \vec{B}_1 \approx B_1 \mu(-\sin\omega t, \cos\omega t, 0)$

$$\vec{\mu}_\perp \times \vec{B}_0 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \mu_x & \mu_y & 0 \\ 0 & 0 & B_0 \end{vmatrix} = \hat{x}\,\mu_y B_0 - \hat{y}\,\mu_x B_0$$

$$\Rightarrow \frac{d\vec{\mu}_\perp}{dt} = \gamma\,\vec{\mu} \times \vec{B} - \frac{\vec{\mu}_\perp}{T_2} = \gamma\,\vec{\mu} \times (\vec{B}_0 + \vec{B}_1) - \frac{\vec{\mu}_\perp}{T_2}$$
$$= \gamma\,\vec{\mu}_\perp \times \vec{B}_0 + \gamma\,\vec{\mu} \times \vec{B}_1 - \frac{\vec{\mu}_\perp}{T_2}$$

$$\frac{d\mu_x}{dt} = \gamma \mu_y B_0 - \gamma \mu B_1 \sin\omega t - \frac{\mu_x}{T_2} = -\omega_0 \mu_y - \gamma B_1 \mu \sin\omega t - \frac{\mu_x}{T_2}$$

$$\frac{d\mu_y}{dt} = -\gamma \mu_x B_0 + \gamma \mu B_1 \cos\omega t - \frac{\mu_y}{T_2} = \omega_0 \mu_x + \gamma B_1 \mu \cos\omega t - \frac{\mu_y}{T_2}$$

$$\frac{d}{dt}(\mu_x - i\mu_y) = -\omega_0 \underbrace{(\mu_y + i\mu_x)}_{i(\mu_x - i\mu_y)} - \gamma B_1 \mu \underbrace{(\sin\omega t + i\cos\omega t)}_{+i(\cos\omega t - i\sin\omega t)} - \frac{\mu_x - i\mu_y}{T_2}$$

let $\mathcal{M} = \mu_x - i\mu_y \Rightarrow$

$$\frac{d\mathcal{M}}{dt} + i\omega_0 \mathcal{M} + \frac{\mathcal{M}}{T_2} = -i\gamma B_1 \mu e^{-i\omega t}$$

$\Rightarrow$ let $\mathcal{M}(t) = \mathcal{M}_0 e^{-i\omega t}$ be the <u>partial</u> solution

$$\Rightarrow -i\omega \mathcal{M}(t) + i\omega_0 \mathcal{M}(t) + \frac{\mathcal{M}(t)}{T_2} = -i\gamma B_1 \mu e^{-i\omega t}$$

$$\left(i\omega_0 - i\omega + \frac{1}{T_2}\right) \mathcal{M}_0 e^{-i\omega t} = -i\gamma B_1 \mu e^{-i\omega t}$$

$$\mathcal{M}_0 = \frac{-i\gamma B_1 \mu}{-i\omega + i\omega_0 + \frac{1}{T_2}} = \frac{\gamma B_1 \mu}{\omega - \omega_0 + \frac{i}{T_2}}$$

$$\mathcal{M}(t) = \mathcal{M}_0 e^{-i\omega t} = \frac{\gamma B_1 \mu e^{-i\omega t}}{\omega - \omega_0 + \frac{i}{T_2}} = \frac{\gamma B_1 \mu e^{-i\omega t - i\alpha}}{\sqrt{(\omega-\omega_0)^2 + \frac{1}{T_2^2}}}$$

Write $\omega - \omega_0 + \frac{i}{T_2} = \sqrt{(\omega-\omega_0)^2 + \frac{1}{T_2^2}}\, e^{i\alpha}$

$$\Rightarrow \frac{\sin\alpha}{\cos\alpha} = \frac{1}{T_2(\omega-\omega_0)} = \tan\alpha.$$

$$\Rightarrow \mu_x(t) = \frac{\gamma B_1 \mu \cos(\omega t + \alpha)}{\sqrt{(\omega-\omega_0)^2 + \frac{1}{T_2^2}}}$$

$$\mu_y(t) = \frac{\gamma B_1 \mu \sin(\omega t + \alpha)}{\sqrt{(\omega-\omega_0)^2 + 1/T_2^2}}$$

resonance at $\omega = \omega_0$ when the amplitude becomes $\gamma B_1 T_2 \mu$.

and $\tan\alpha = \infty \Rightarrow \alpha = \pi/2$.

$$\mathcal{M}(t) = \gamma B_1 T_2 \mu e^{-i\omega t - i\pi/2}$$

<u>General</u> solution: $\mathcal{M}(t) = \mathcal{M}(0)\, e^{-i\omega_0 t} e^{-t/T_2}$

decay to 0 for $t \gg T_2$.

Our approximation is based on (1) $B_1 \ll B_0$, (2) $\mu_{x,y} \ll \mu_z$.
But at the resonance $\mu_{x,y} \sim \gamma B_1 T_2 \mu \Rightarrow$ need $\gamma B_1 T_2 \ll 1$.

We can do better and get into non-linear regime
Let $\vec{B} = (B_1 \cos\omega t, B_1 \sin\omega t, B_0)$ and omit relaxation $T_{1,2} \to \infty$:

$$\begin{cases} \dfrac{d\mu_x}{dt} = -\omega_0 \mu_y + \omega_1 \sin\omega t\, \mu_z \\ \dfrac{d\mu_y}{dt} = \omega_0 \mu_x - \omega_1 \cos\omega t\, \mu_z \\ \dfrac{d\mu_z}{dt} = -\omega_1 \sin\omega t\, \mu_x + \omega_1 \cos\omega t\, \mu_y \end{cases}$$

$\omega_0 = -\gamma B_0$
$\omega_1 = -\gamma B_1$

$\mu_x^2 + \mu_y^2 + \mu_z^2 = \mu_0^2 \Rightarrow$ only 2 equations are independent.

$\mu_x = \mu_0 \sin\theta \cos\varphi$, $\mu_y = \mu_0 \sin\theta \sin\varphi$, $\mu_z = \mu_0 \cos\theta$

$$\begin{cases} (1) \quad \cos\theta \cos\varphi \dfrac{d\theta}{dt} - \sin\theta \sin\varphi \dfrac{d\varphi}{dt} = -\omega_0 \sin\theta \sin\varphi + \omega_1 \sin\omega t \cos\theta \\ (2) \quad \cos\theta \sin\varphi \dfrac{d\theta}{dt} + \sin\theta \cos\varphi \dfrac{d\varphi}{dt} = +\omega_0 \sin\theta \cos\varphi - \omega_1 \cos\omega t \cos\theta \end{cases}$$

$-\sin\varphi \cdot (1) + \cos\varphi \cdot (2)$:

$$\sin\theta \frac{d\varphi}{dt} = \omega_0 \sin\theta - \omega_1 \underbrace{(\sin\omega t \sin\varphi + \cos\omega t \cos\varphi)}_{\cos(\varphi - \omega t)} \cos\theta$$

$\Rightarrow$ get $\sin\theta \dfrac{d\varphi}{dt} = \omega_0 \sin\theta - \omega_1 \cos\theta \cos(\varphi - \omega t)$. (4)

(3) $-\sin\theta \dfrac{d\theta}{dt} = -\omega_1 (\sin\omega t \cos\varphi - \cos\omega t \sin\varphi) \sin\theta$

$\Rightarrow \dfrac{d\theta}{dt} = -\omega_1 \sin(\varphi - \omega t)$. (5)

Let $\tilde{\varphi} = \varphi - \omega t \Rightarrow \dfrac{d\tilde{\varphi}}{dt} = \dfrac{d\varphi}{dt} - \omega$ $\quad$ $\tilde{\varphi}$ is angle in the frame $\tilde{K}$ rotating with ω, external frequency.

$$\Rightarrow \quad \frac{d\theta}{dt} = -\omega_1 \sin\tilde{\varphi}$$

$$\frac{d\tilde{\varphi}}{dt} = -\Delta\omega - \omega_1 \cot\theta \cos\tilde{\varphi} \quad , \quad \Delta\omega = \omega - \omega_0$$

Note: if $\tilde{\varphi} = \cancel{\text{const}}$ const $\Rightarrow 0 = -\Delta\omega - \omega_1 \cot\theta \cos(\text{const})$.

$\Rightarrow \theta = $ courtant too. $= \theta_0$.

But then $\dfrac{d\theta}{dt} = 0 \Rightarrow \tilde{\varphi} = 0 \Rightarrow \underline{\text{const} = 0}$.

Then $\tilde{\varphi} = 0$, $\cot\theta_0 = -\dfrac{\Delta\omega}{\omega_1}$

$$\Rightarrow \frac{\cos\theta_0}{\sqrt{1-\cos^2\theta_0}} = -\frac{\Delta\omega}{\omega_1} \Rightarrow \frac{\cos^2\theta_0}{1-\cos^2\theta_0} = \left(\frac{\Delta\omega}{\omega_1}\right)^2$$

$$\Rightarrow \left[1 + \left(\frac{\Delta\omega}{\omega_1}\right)^2\right] \cos^2\theta_0 = \left(\frac{\Delta\omega}{\omega_1}\right)^2$$

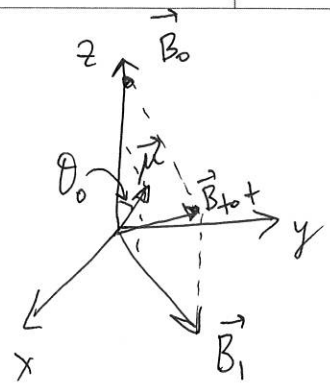
$$\Rightarrow \cos^2\theta_0 = \frac{(\Delta\omega)^2}{\sqrt{(\omega_1^2 + (\Delta\omega)^2)}} \Rightarrow \cos\theta_0 = -\frac{\Delta\omega}{\sqrt{(\Delta\omega)^2 + \omega_1^2}}$$

$$\sin\theta_0 = \frac{\omega_1}{\sqrt{(\Delta\omega)^2 + \omega_1^2}}$$

Here $\Omega = \sqrt{(\Delta\omega)^2 + \omega_1^2}$ = Rabi frequency.

$\Rightarrow$ In $\tilde{K}$ $\vec{\mu} = \mu_0 \tilde{n}_0$, $\tilde{n}_0 = (\sin\theta_0, 0, \cos\theta_0)$

in K $\vec{\mu}$ rotates frequency ω on a cone of angle θ_0.



What if $\theta \neq \theta_0$? Expand in small $\theta - \theta_0$:

$$\cot\theta = \cot(\theta - \theta_0 + \theta_0) \approx \cot\theta_0 + (\theta-\theta_0)(\cot\theta_0)' = \cot\theta_0 - \frac{\theta-\theta_0}{\sin^2\theta_0}$$

$$(\cot\theta)' = \left(\frac{\cos\theta}{\sin\theta}\right)' = \frac{-\sin\theta}{\sin\theta} - \frac{\cos\theta\cdot\cos\theta}{\sin^2\theta} = -\frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta} = -\frac{1}{\sin^2\theta}$$

Also expand in $\tilde{\varphi}$: $\sin\tilde{\varphi} \to \tilde{\varphi}$, $\cos\tilde{\varphi} \to 1$

$$\frac{d\theta}{dt} = -\omega_1 \sin\tilde{\varphi} \to -\omega_1\tilde{\varphi}$$

$$\frac{d\tilde{\varphi}}{dt} = -\Delta\omega - \omega_1 \cot\theta \cos\tilde{\varphi} \to -\Delta\omega - \omega_1\left(\cot\theta_0 - \frac{\theta-\theta_0}{\sin^2\theta_0}\right) =$$

$$= -\Delta\omega - \omega_1 \underbrace{\cot\theta_0}_{-\frac{\Delta\omega}{\omega_1}} + \frac{\omega_1}{\sin^2\theta_0}(\theta-\theta_0) = \frac{\Omega^2}{\omega_1}(\theta-\theta_0)$$

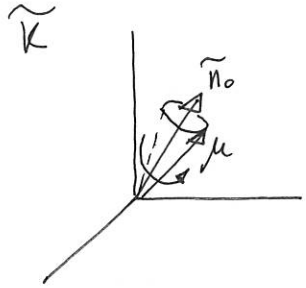
Then
$$\left.\begin{aligned} \frac{d^2\theta}{dt^2} &= -\omega_1\frac{d\tilde{\varphi}}{dt} \\ \frac{d\tilde{\varphi}}{dt} &= \frac{\Omega^2}{\omega_1}(\theta-\theta_0) \end{aligned}\right\} \Rightarrow \frac{d^2\theta}{dt^2} = -\Omega^2(\theta-\theta_0)$$

$\theta(t)$ oscillates harmonically with $\Omega = \sqrt{\omega_1^2 + (\Delta\omega)^2} =$

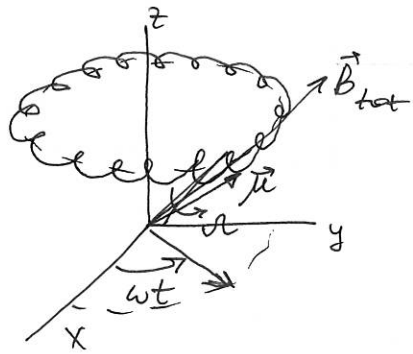
$$= \sqrt{\omega_1^2 + (\omega-\omega_0)^2}$$

In quantum mechanics this is known as <u>Rabi oscillations</u>

In the rotating frame $\tilde{K}$ $\vec{\mu}$ precesses with Rabi frequency Ω around the stationary direction $\tilde{n}_0$.



In lab frame K this Rabi rotation is superimposed over the precession around $\hat{z}$-axis with frequency ω (of the driving field $\vec{B}_1$)



Classically, this is <u>nutation</u>.